MonkeyTax — Visual Case Gallery

Every observable state of the extension, captured end-to-end in a real Chromium/Edge browser with `dist/` loaded unpacked via `node scripts/capture-screenshots.mjs`. Badge content is produced by the actual content script running against per-platform fixture DOMs served at the `gemini.google.com`, `chatgpt.com`, and `claude.ai` origins (including the Gemini `You said …` duplication quirk), so each screenshot reflects the true extraction → classification → calculation → render pipeline with **2025/2026-calibrated coefficients** (output tokens carry a ~5× prefill-to-decode energy ratio, and every figure is rounded into everyday visuals — sips, shot glasses, phone charges).

> Regenerate everything: `npm run build && node scripts/capture-screenshots.mjs` Refresh the PDF from this document: `npm run export:showcase`

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1. Intent tiers — collapsed icon widgets

Every badge is a **centered pill** of compact visual widgets, collapsed by default and free of walls of text (copy rule: zero em/en dashes everywhere). INFORMATIVE shows `[💧 ~2 sips] [💡 4m lamp] [⚡ 2× web search] [▾]`. Roast tiers are **instant regret**: warning-red styling and the full punchline on one line, visible with zero clicks — `[🔥] [Headline. tail sentence…] [N× search chip] [▾]` — ending in a conversational money line, then `[▾]` (the drawer adds only numbers, never a second helping of the joke). Chevron toggles the drawer (§2).

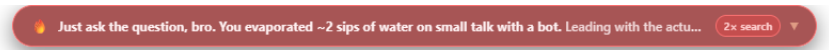
1.1 `POLITE` — roast tier

Prompt: `Thanks` — the red roast pill delivers the whole joke instantly: *"Being polite to a toaster: You just evaporated ~2 sips of water for good manners. A thumbs-up emoji on your keyboard would have cost zero energy."*



1.2 `GREETING` — roast tier

Prompt: `Good morning!` — zero-question openers roasted on sight: *"Woke up a billion-dollar server cluster just to say hi with zero question attached."*



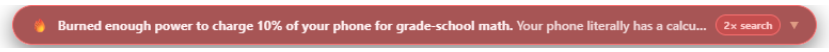
1.3 `LAZY_RETRY` — roast tier

Prompt: `fix it` — zero-context rage-commands ("try again", "???") are caught only after the technical guardrails pass, e.g. *"'Fix it' is not a prompt. Making the AI guess what broke burned 2 min of lamp power."*



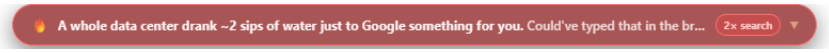
1.4 `SIMPLE_MATH` — roast tier

Prompt: `tell me how much is 7777+998` — natural-language math is caught by the classifier ("what is…", "calculate…", "tell me how much is…", "solve…" prefixes all fold into SIMPLE_MATH). Collapsed pill shows the full punchline, e.g. "*Burned enough power to charge 10% of your phone for grade-school math. Your phone literally has a calculator app sitting right there.*"



1.5 `QUICK_SEARCH` — roast tier

Prompt: `capital of france` — ultra-short factual lookup roasted with lines like "*A whole data center drank ~2 sips of water just to Google something for you.*"



1.6 `INFORMATIVE` — neutral plain-language readout (~90 % of traffic)

Prompt: `Write a TypeScript debounce function` — pure icon widgets: `💧 ~2 sips` `💡 4m lamp` `⚡ 2× web search` with no prose at all.



1.7 `INFORMATIVE` + extended thinking — 2.0× reasoning multiplier

Prompt: `Prove that the sum of two even numbers is even` (reasoning UI present) — same template, every figure carries `REASONING_MULTIPLIER = 2.0`.



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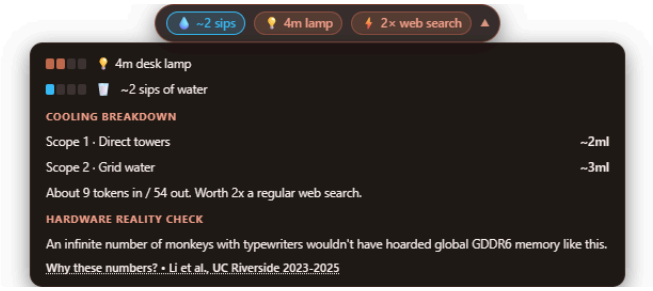
2. The expandable accordion drawer

Clicking a badge opens its details; clicking again closes them. Gemini and ChatGPT use an inline accordion. Claude uses a bounded native popover instead: its virtualized message rows keep exactly the same height, and opening details never calls `scrollIntoView`. The badge is inserted before the message action bar, and the drawer is positioned above the actual composer boundary. Escape, outside clicks, scrolling, or resizing dismiss it. Its document listeners exist only while open. The drawer leads with **in-line segment meters** and icon visuals, then the details the collapsed pill hides:

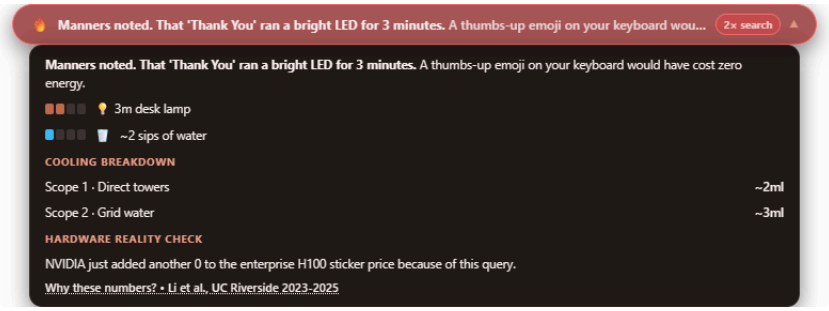
- **Energy meter** — 4-segment battery scaled against the 0.3 Wh web search, tagged `💡 Nm desk lamp` (or laptop-load seconds).
- **Water meter** — 4-segment glass bar with discrete `🥛` icons and the everyday visual (`~2 sips of water`).
- **Cooling breakdown** — *Scope 1 · Direct towers* (on-site evaporation: `2 ml + 2 ml per 1k output tokens`) and *Scope 2 · Grid water* (off-site thermoelectric: `5.0 ml per Wh`).
- **Plain-language token line** (neutral tiers) and a rotating line from the Reddit hardware-crisis roast bank. Roast tiers open with the **complete punchline, fully wrapped**: the collapsed pill can ellipsize

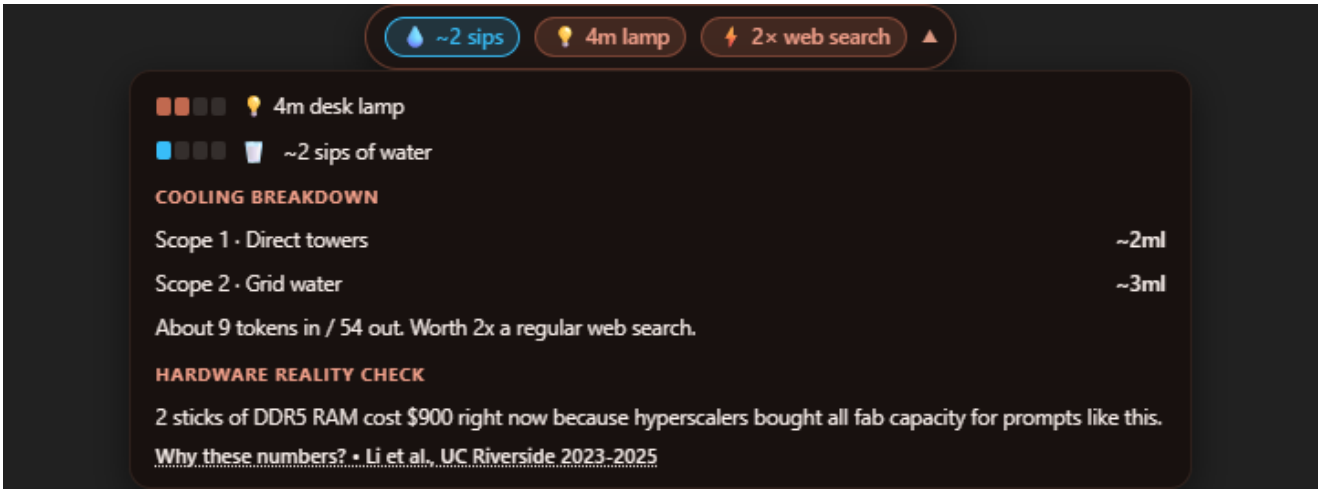
on narrow viewports, so the drawer guarantees the whole joke is readable after the click.

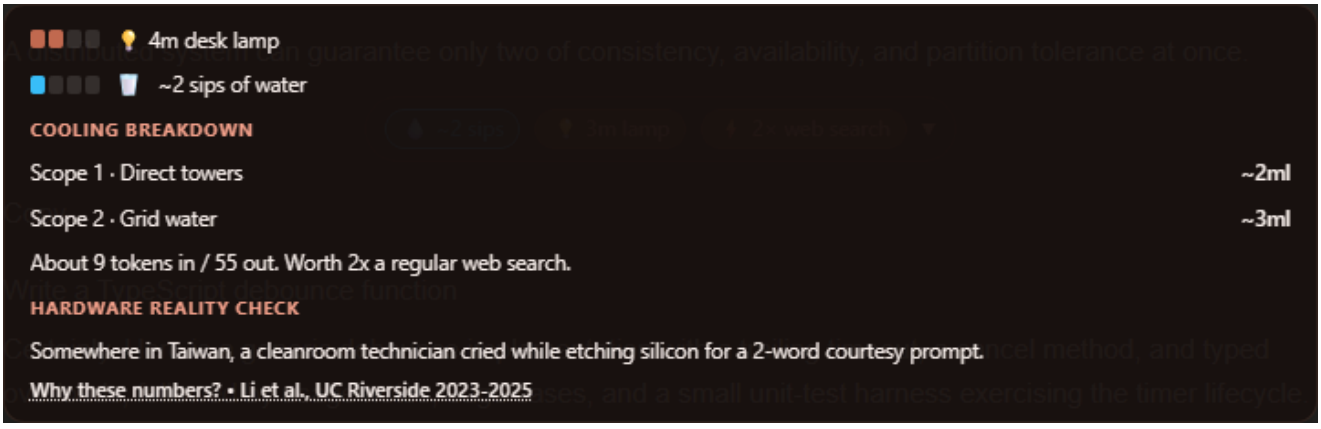
- **Citation link** — opens the scientific modal (§4).



Roast drawer after click (full punchline, wrapped):

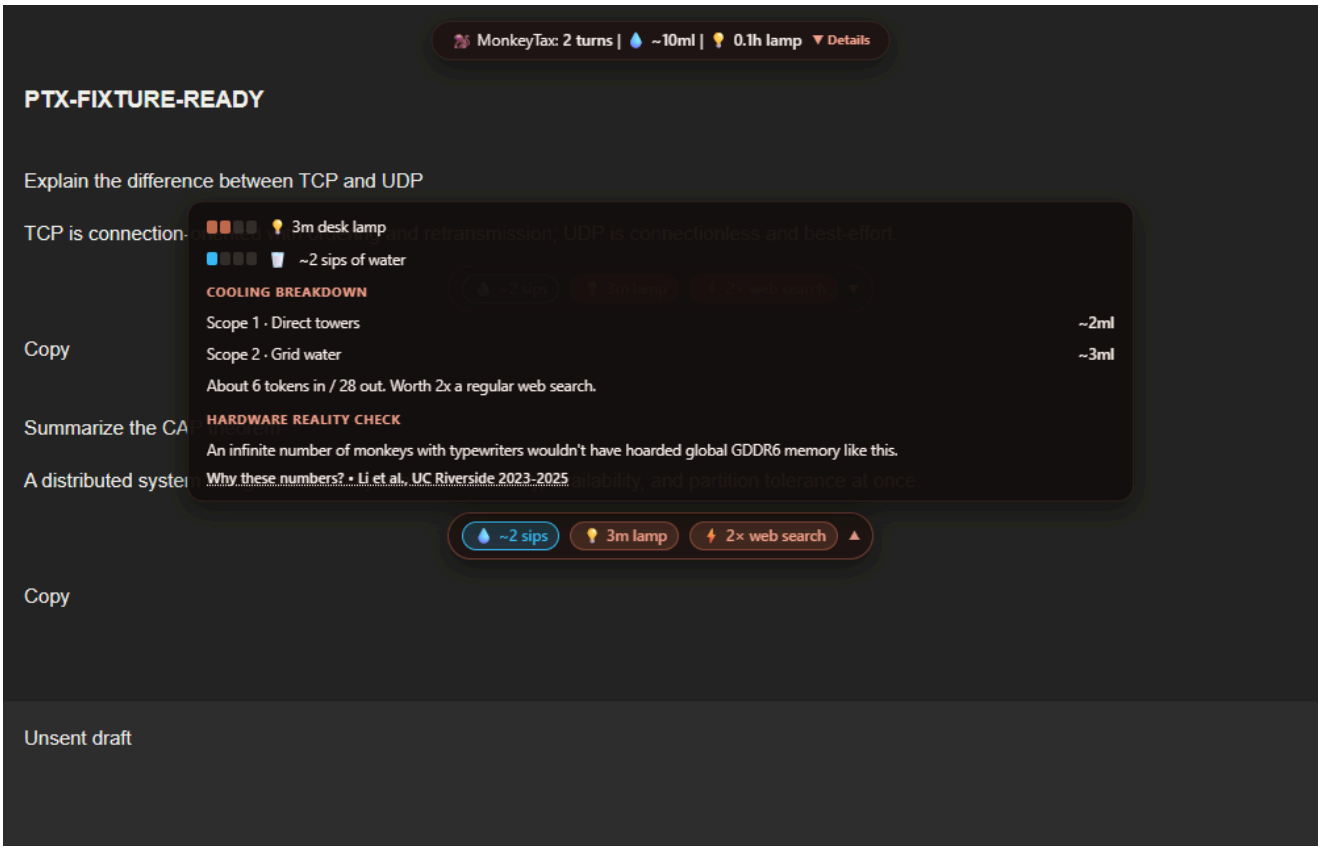


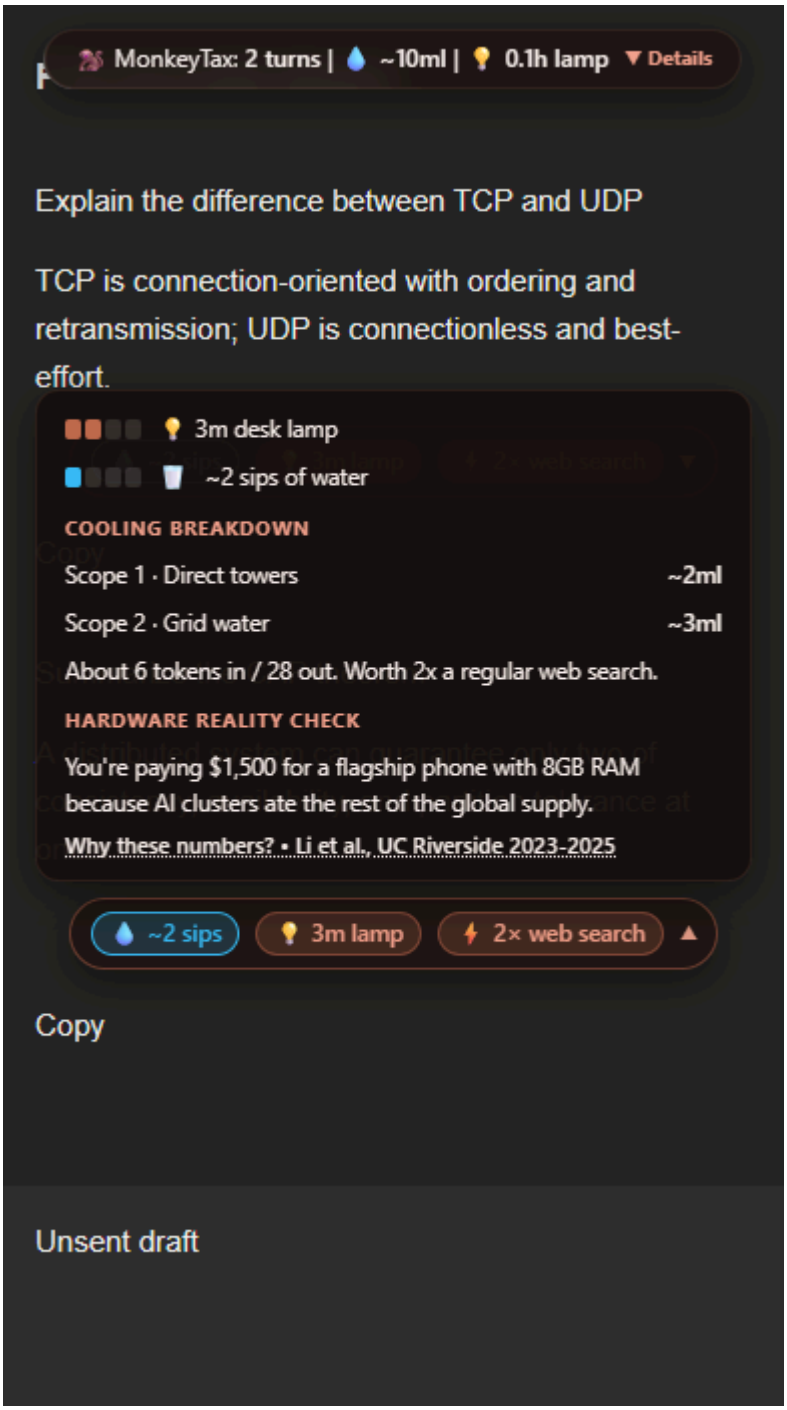




Fixed-composer regressions at desktop, compact-window, and mobile sizes assert unchanged scroll position, scroll height, and badge height on open; a clear composer boundary; dismissal on Escape and scroll; and preservation of an unsent composer draft. The collapsed Claude badge also uses a local

stacking level rather than the extension's global overlay z-index. An additional regression deliberately scrolls the pill behind the fixed composer and verifies that the composer wins pointer hit-testing and receives input focus. The test scroll panel is not a fixed-position stacking context, which would otherwise mask the overlap seen on the real site.


MonkeyTax: 2 turns | ~10ml | 0.1h lamp Details
PTX-FIXTURE-READY
Explain the difference between TCP and UDP
TCP is connection-
3m desk lamp
~2 sips of water
COOLING BREAKDOWN
Scope 1 · Direct towers ~2ml
Scope 2 · Grid water ~3ml
About 6 tokens in / 28 out. Worth 2x a regular web search.
HARDWARE REALITY CHECK
An infinite number of monkeys with typewriters wouldn't have hoarded global GDDR6 memory like this.
Why these numbers? • Li et al., UC Riverside 2023-2025
Copy
Summarize the CA
A distributed syste
~2 sips
3m lamp
2× web search
Copy
Unsent draft



3. Thread-total dock: expandable dashboard + deep-history scanner

Opening a **pre-existing** conversation (`chatgpt.com/c/…` , `gemini.google.com/app` , `claude.ai/chat/…`) mounts a translucent dark-glass dock **pinned to the top center** showing `📝 Thread: N turns | 💧 ~Xml | 💡 Nh lamp | ▼ Details` .

Clicking ▼ **Details** expands an interactive card: a 4-stat dashboard (water with a mini bottle-fill icon, power + lamp hours, web-search equivalence, microwave blast seconds below 20 Wh or continuous phone screen-on time above it) plus a ⚡ **Scan Entire History** button. Chat apps virtualize their message

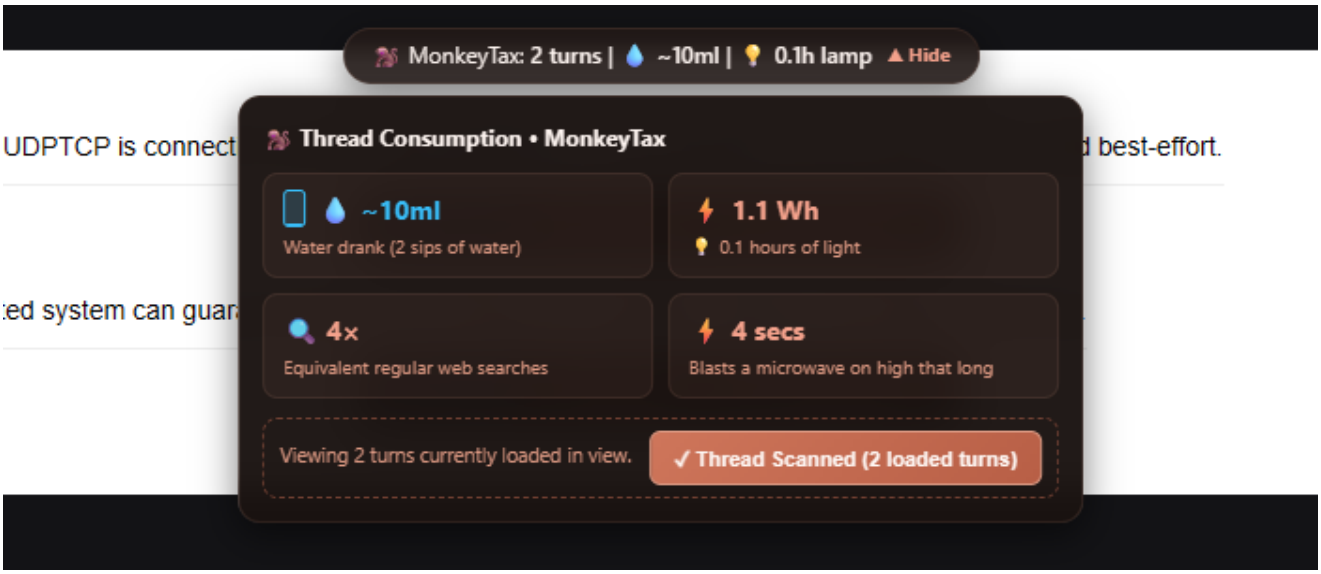
lists, so only a window of messages may exist in the DOM. The scanner moves upward in overlapping viewport steps every 250 ms, retains observed totals by stable response ID, rediscovers replaced scroll panels, and waits for three seconds of quiet at the top. It leaves the chat at the top rather than restoring the old scroll position. Its 60-second bound reports a limit instead of claiming completion when loading has not settled. Gemini uses a seven- second quiet window and re-arms its top-edge loader every 1.5 seconds. Its Material spinner check requires the loader to intersect the chat viewport, so an offscreen spinner cannot hold the scan open forever. The button reports `✓ Thread Scanned (N loaded turns)`; unloaded or inaccessible history cannot be guaranteed. The per-conversation ID cache resets on navigation and prevents recycled message nodes from duplicating thread totals. Discovering old messages does not create new daily receipts.

Gemini's append-only `infinite-scroller.chat-history` uses a separate top-edge pagination retry: the scan reaches the top, waits for Material history-loading spinners that intersect the chat viewport, and moves beyond the top viewport before returning to trigger another page. Offscreen spinners do not count as active loading. The longer quiet window accommodates network pauses between page fetches. A Gemini-specific regression keeps an offscreen spinner mounted and requires three separately triggered pages to load from one scan-button click without creating daily receipts. This final change was validated in the local browser harness; the live Edge bridge timed out before the updated code could be checked against the signed-in Gemini conversation.

Regression coverage includes three delayed history batches and a 14-message virtualized transcript on both Claude and ChatGPT. Each single-click scan must retain all seven virtualized turns, finish at scroll position zero, preserve unrelated sidebar position and inline scroll styles, and leave daily prompt counts unchanged.



Expanded dashboard after a full-history sweep:



3.5 Adaptive theme engine (live dark ⇄ light)

The MonkeyTax brand palette uses terracotta `#C1694F` and soft peach `#DF9B84`: roasted espresso surfaces in dark mode, warm cream surfaces and espresso text in light mode. Water telemetry remains blue; roast warnings remain red. Canonical `--mtx-*` tokens back the existing `--ptx-*` aliases, preserving component and data-attribute compatibility. The popup uses a local Twemoji PNG avatar and retains linked CC-BY 4.0 attribution after receipt updates. Toolbar icons are generated at 16/32/48/128 pixels with approximately 8% padding per side.

A theme detector (`src/content/ui/theme-detector.ts`) reads the **real host page** — `dark` / `light` classes on `<html>` / `<body>` , `data-theme` / `data-mode` attributes, inline `color-scheme` , background luminance fallback, OS media query last — and a `MutationObserver` on `<html>` / `<body>` (plus `matchMedia`) flips every mounted surface **without a page reload**: badge pills, drawers, the thread dock, and the citation modal all consume one CSS custom-property token set bound to `:host([data-ptx-theme=…])` . Roast pills keep a dedicated high-contrast red theme in *both* modes (dark: `#F87171` border; light: `#EF4444` border on `#FEF2F2`). The captures below were produced by toggling the fixture theme mid-session, and the harness asserts every mounted host carries the flipped `data-ptx-theme` attribute.

Thread dock, light mode:



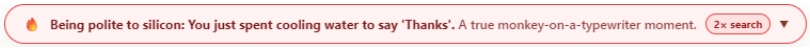
INFORMATIVE pill, light mode:



Drawer expanded, light mode:



Roast pill, light mode (contrast-preserving red tokens):



Production markup support: selectors target ChatGPT's current `data-chatgpt-search-unit-key` values ending in `:user` / `:assistant`, as well as the older `article[data-testid^="conversation-turn-"]` / `.agent-turn` with `.markdown/.prose` bodies and the global stop-button streaming gate; Claude `[data-testid="user-message"]` / `.font-user-message` with backwards-walk user pairing; a generic document-order user⇄response zipper (all three platforms) takes over whenever an adapter's strict selectors see nothing after a redeploy. Sidebar thread switches (SPA `pushState`, no reload) are caught by an 800 ms `location.href` watch that wipes processed-turn tracking, re-seeds history, and rebuilds the dock after the new thread hydrates.

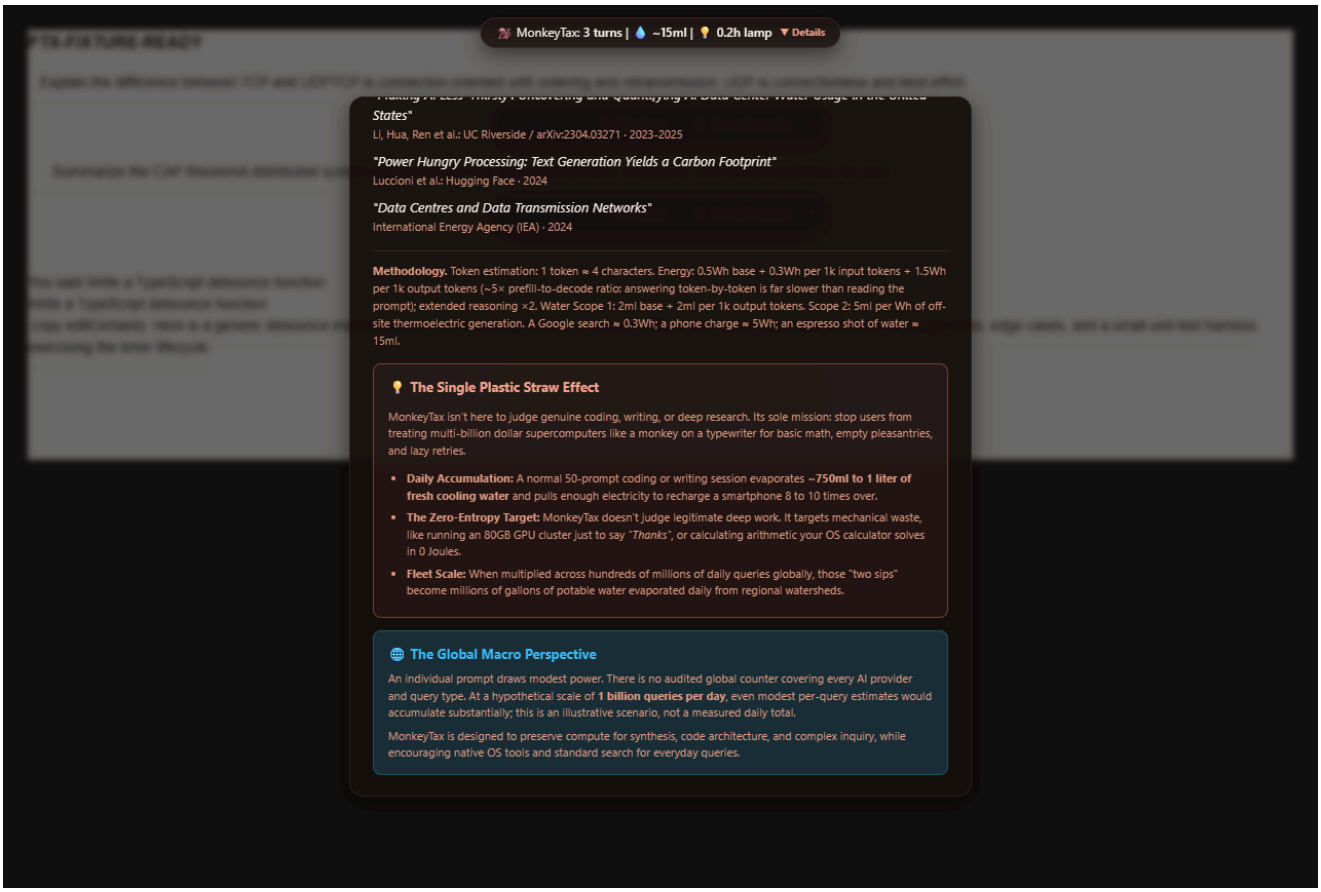
Live Edge testing on 2026-10-26 reproduced the old ChatGPT selector failure and the scanner restoring a bottom scroll position on Claude. The updated extension was exercised against signed-in conversations on both sites, including real drawer clicks and sidebar navigation. History badges are rendered explicitly rather than relying on whether the initial page hydration happened before or after observer startup. On the real 14-message Claude transcript, the virtualized scan retained seven turns at scroll position zero even when only six assistant rows remained mounted. A real ChatGPT scan also reached position zero in a 7,204-pixel conversation with three replies. Real conversation contents and screenshots are not included in this fixture gallery.

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4. Scientific citation modal

Opened via the **"Why these numbers? → Li et al., UC Riverside 2023–2025"** link inside the drawer. Renders the Scope 1 / Scope 2 split, the full citations (UC Riverside `arXiv:2304.03271` / `arXiv:2506.07146`, Hugging Face 2024, IEA 2024), and the methodology constants including the **~5× prefill-to-decode ratio**, the 2025/2026-calibrated coefficients (0.5 Wh base, 0.3/1.5 Wh per 1k input/output tokens, 5.0 ml/Wh grid water), and the 2.0× reasoning multiplier. The highlighted **"The Single Plastic Straw Effect"** card answers *why do these numbers look small per prompt*: daily accumulation (a 50-prompt session ≈ 1 liter of cooling water), the zero-entropy target (mechanical waste like `"Thanks"`, not deep work), and fleet-scale watershed impact. Dismisses on ×, backdrop, or `Escape`.

The **Global Macro Perspective** panel avoids stating an unaudited global query count as fact. It presents 1 billion queries/day only as a hypothetical scale scenario and explains that no audited cross-provider daily counter is available.



5. Per-platform rendering

One INFORMATIVE turn on each supported platform — identical centered pill and metric chip everywhere; the platform adapters differ only in their extraction selectors.

Gemini:



ChatGPT:



Claude:

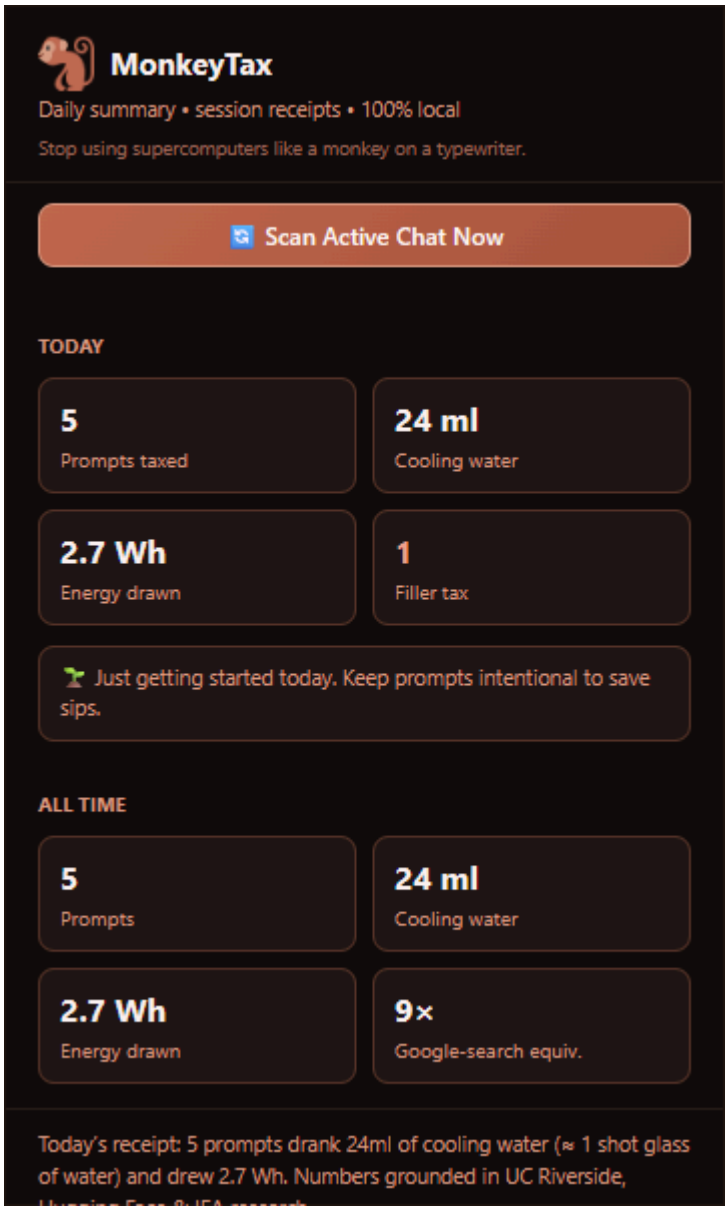


6. Toolbar popup — session receipts

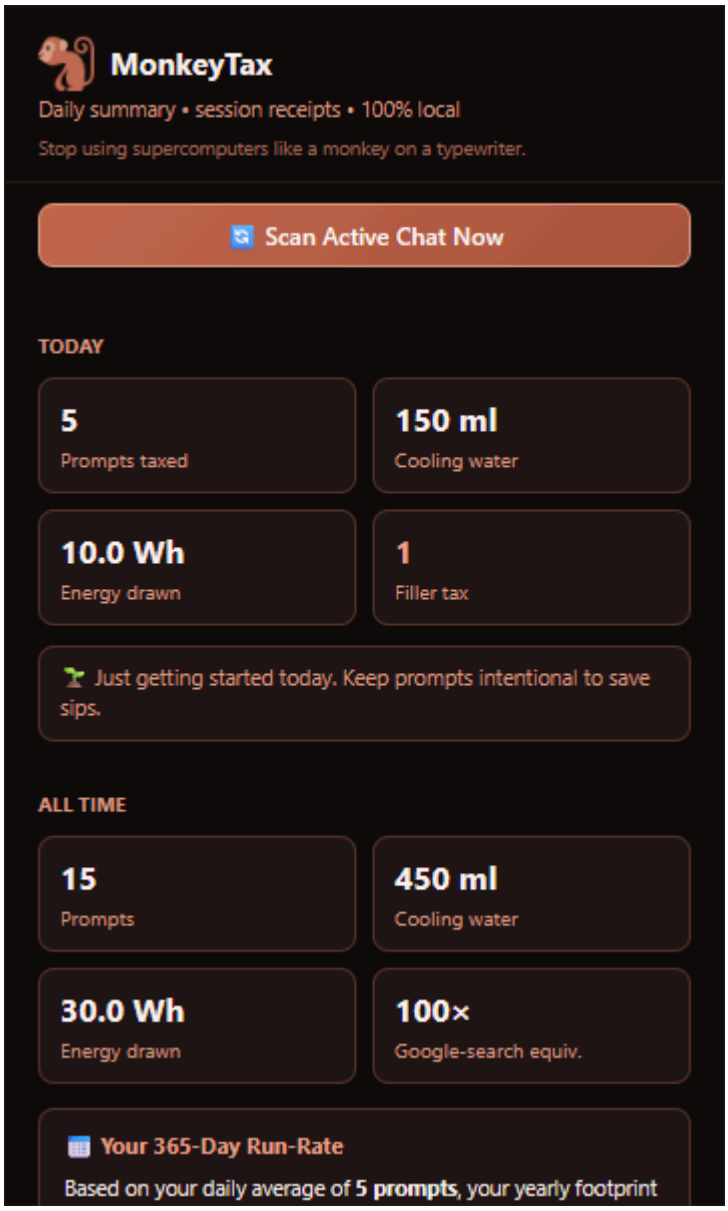
6.1 Populated

Rolling per-day and all-time aggregates written to `chrome.storage.local` by the service worker: prompts taxed, cooling water (Scope 1 + 2), energy, filler-tax count (politeness, greetings, lazy retries), and Google-search equivalent. Manual thread scans persist once per platform/conversation fingerprint; repeated clicks do not add the same thread twice. Existing `days` storage migrates to `dailyHistory` while remaining mirrored for compatibility. The receipt footer now speaks in the same human visuals as the pills (`≈ a full reusable bottle` , not espresso shots). Under the daily grid, a contextual **Scale Reality Check** badge escalates with volume: 🌱 under 10 prompts, ⚡ phone-charge equivalence at 10–29 (as pictured), or 🚰 water-bottle count at 30+. At the top sits 🔄 **Scan Active Chat Now**: it messages the content script in the active tab (`PTX_FORCE_SCAN`), which force-scans every pair currently in the DOM, injects badges on turns the live observer never saw, and answers with the scanned totals (`✓ Scanned N turns (Xml water, Y Wh)`). If the tab predates an extension reload it self-heals via `chrome.scripting` injection, and falls back to "Please refresh this tab once to activate MonkeyTax."

At 15 all-time prompts, **Your 365-Day Run-Rate** projects the user's own average daily water and energy use across 365 days and active days. It does not multiply the user's numbers by an arbitrary population factor.

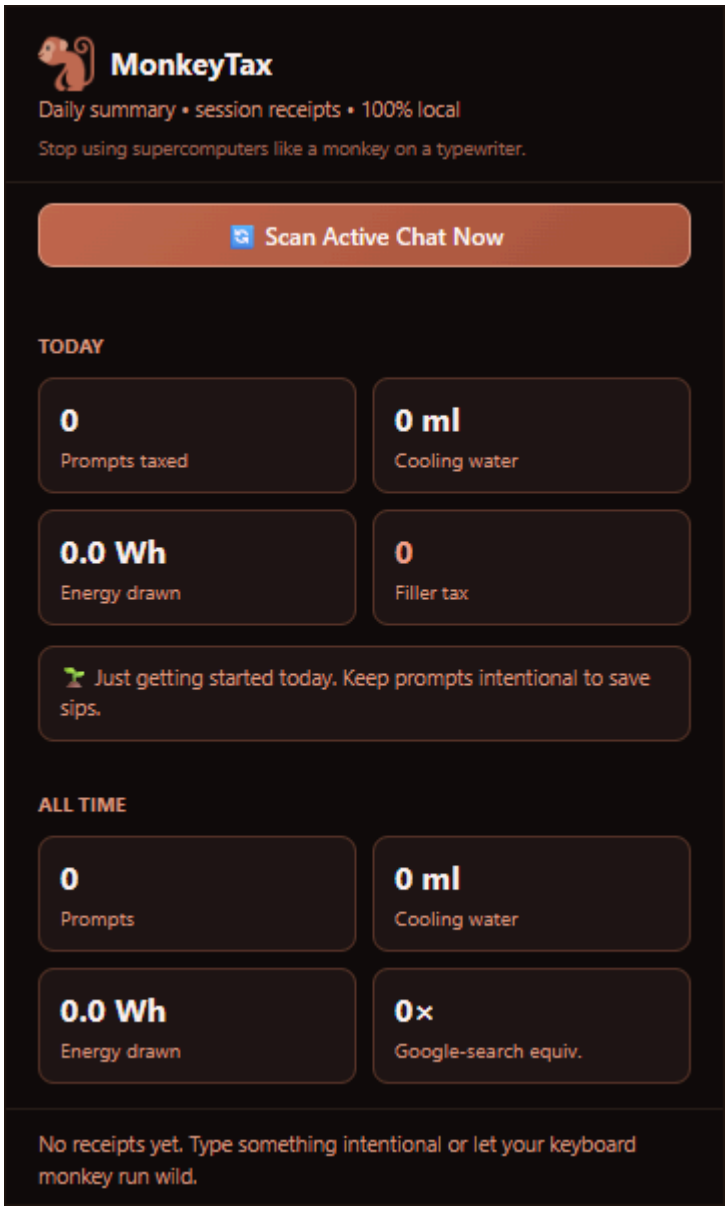


Annual run-rate at the 15-prompt threshold:



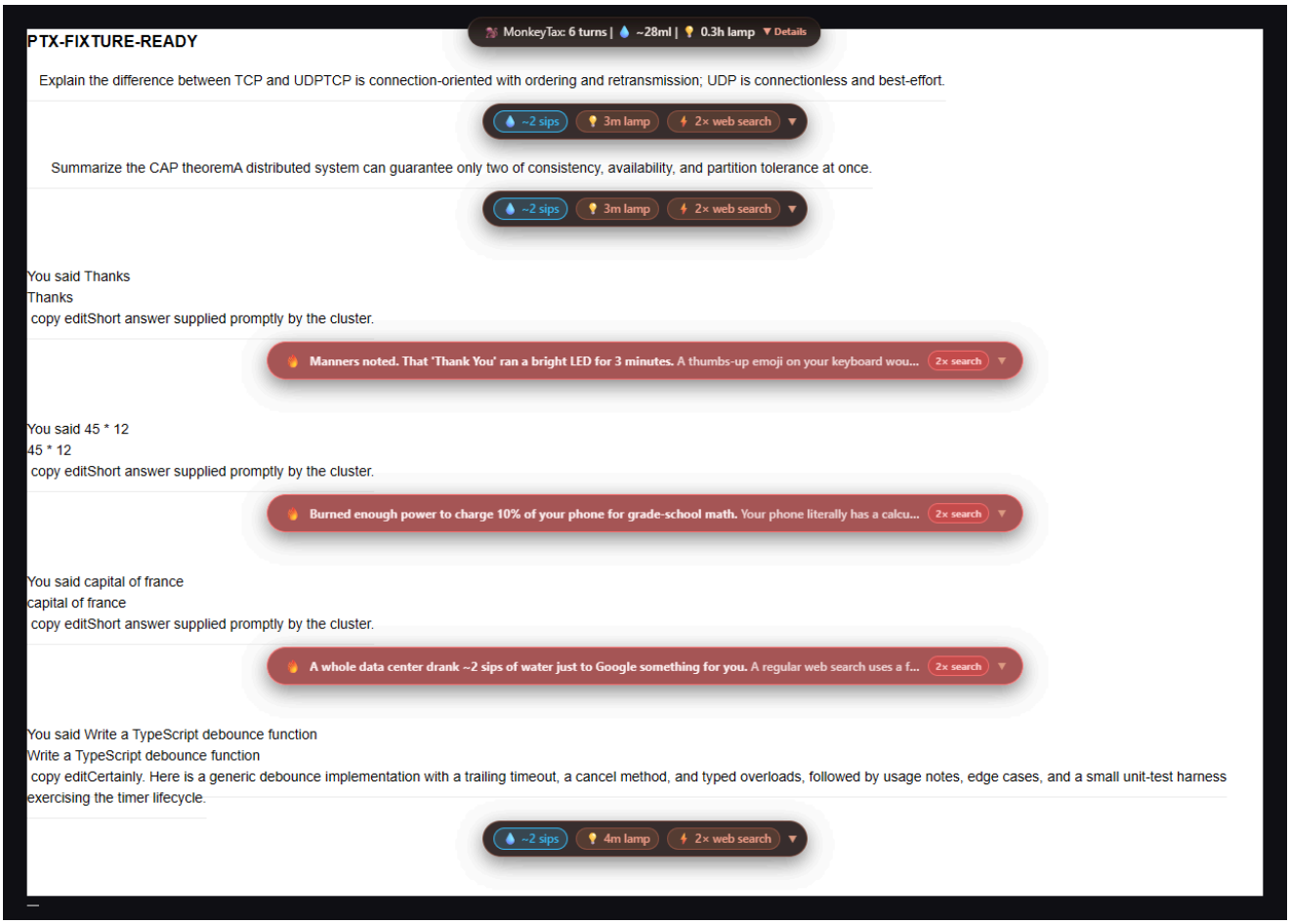
6.2 Zero state

Before any prompt has been recorded: zeroed stat grid and the "no receipts yet" footer.



7. Multi-tier conversation overview

One Gemini thread: the seeded history folded into the top dock (6 turns total), then four live turns — one per intent tier — each badge centered under its own assistant response; roast tiers red, neutral tiers terracotta with blue water telemetry.



8. Pipeline behaviour proven by these captures

Each screenshot was taken only after the harness asserted the badge's `data-ptx-tier` / `data-ptx-roast` attributes and the drawer's `data-ptx-open` flag, and the thread dock's `data-ptx-turns` / `data-ptx-water-ml` / `data-ptx-energy-wh` attributes per platform (the sanctioned inspection surface — badges live in **closed** shadow roots by design).

Behaviour	Evidence
Output-weighted math (~5× prefill-to-decode; Scope 1 + Scope 2 water)	drawer screenshots §2, telemetry in `cases.json`
Gemini `You said X` accordion duplication collapsed	`sanitized: "Thanks"` telemetry line, every case
Leaked `copy edit` action labels + zero-width glyphs stripped	same telemetry line

Behaviour	Evidence
Roast tiers never misfire on genuine work	25-case false-positive matrix (`tests/classifier-edge-cases.spec.ts` + `scripts/test-edge-cases.mjs` , 25/25 green): "Thanks, now refactor this function…", "What is the time complexity of quicksort?", "Prove that 2 + 2 = 4 using Peano axioms", "Fix this TypeScript regex…", "Why is 0.1 + 0.2 !== 0.3 in JavaScript?" all stay `INFORMATIVE`
Full roast punchline visible on the COLLAPSED pill	`data-ptx-pill-chars >= 35` + `data-ptx-pill-theme="red"` asserted pre-click for every roast case; ellipsized tails always readable in the drawer (§2)
Scan button works under REAL mouse clicks	capture clicks the closed-shadow button at its published `data-ptx-scan-btn-rect` coordinates and asserts `data-ptx-scanned` matches `data-ptx-turns`
Badge fires only after streaming settles (800 ms quiet)	every screenshot shows final-length responses
Existing-chat scanning of pre-seeded history	§3 dock turn counts per platform
No double-counting between history scan and live turns	live assertion `turns = seeded + 1` per platform
Drawer toggle works on every platform	`data-ptx-open="true"` asserted before each §2 shot
Theme flips live, every surface	harness toggles `<html class>` mid-capture and asserts `data-ptx-theme` on **all** mounted hosts before every §3.5 shot
Expanded dock dismisses on outside click	capture-fold before subsequent badge clicks; `composedPath` -guarded document listener

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9. State matrix (completeness map)

#	State	Tier / flag	Screenshot
1	Politeness filler (collapsed roast)	`POLITE`	§1.1
1b	Zero-question greeting	`GREETING`	§1.2
1c	Zero-context retry / rage-command	`LAZY_RETRY`	§1.3

#	State	Tier / flag	Screenshot
2	Trivial arithmetic, bare or natural-language (`what is 45 * 12` , `calculate 120/4` , `tell me how much is 7777+998` , `solve 9*9` , `45 times 12` , `what is 20% of 450`)	`SIMPLE_MATH`	§1.4
3	Ultra-short lookup (`capital of france` , `weather in bengaluru` , `tokyo time` , `define ephemeral`)	`QUICK_SEARCH`	§1.5
4	Genuine work, collapsed	`INFORMATIVE`	§1.6
5	Genuine work + thinking ×2.0	`INFORMATIVE`	§1.7
6	Drawer expanded (segment meters + Reddit roast)	any	§2 (×3 platforms)
7	Thread dock pill on existing chat	—	§3 (×3 platforms)
7b	Thread dashboard + deep scanner	—	§3
8	Citation modal + Plastic Straw explainer	—	§4
9	Centered rendering per platform	`INFORMATIVE`	§5 (×3)
10	Popup populated / zero (incl. manual Scan button)	—	§6
11	Multi-tier thread + dock together	mixed	§7
12	Dock + pill + drawer + roast pill in LIGHT theme	flipped live	§3.5

Machine-readable run record: `screenshots/cases.json` .